Business Model Canvas

Team / solution: DABAB (ضباب) · AI weather-disruption copilot for ATC Challenge: TDRA Hackathon 2026 · #2 GCAA

7 · Key partners *Who must help us make this real?*

- GCAA: regulator, sponsor, eAIP data
- dans: Dubai air navigation provider
- Dubai Airports: AODB, stands, A-CDM
- Airlines: Emirates, flydubai
- NCM: weather data
- Digital Dubai / Data.Dubai: open flight data
- TDRA: programme and funding route

5 · Key activities *What must we do, again and again?*

- Keep flight, METAR and stand data feeds live
- Tune optimizer and safety rules with controllers
- Replay every weather day to measure results
- Keep AI explanations grounded and checked

6 · Key resources *People, data, tools we need.*

- Data: Dubai Airports flights, METAR (7 UAE airports), GCAA stand and ILS data
- Tools: optimizer engine, LLM explanations with template fallback, ATC panel
- People: engineer, product lead, ATC advisor

2 · Value proposition *What gets better, for whom, and why would they choose this?*

- In fog, storms or dust, controllers get **ranked recovery plans in seconds** instead of manual juggling: who holds, who diverts where, which stand.
- Each plan shows delay, fuel, CO_2, diversions and risk, with a reason in Arabic and English.
- Controller picks the priority (min delay, min fuel, min diversions, protect connections); safety rules are always enforced.
- **Human always approves.** Every decision is logged.
- One shared picture for air navigation, the airport and airlines.

4 · Relationships *How do users start, and keep, using it?*

- Co-designed with controllers
- Shadow mode first: suggests, they compare
- Approve / modify / reject builds trust; audit log
- Training and a review after each weather event

3 · Channels *Where will users find and use it?*

- GCAA-led pilot at DXB
- Runs as a layer on A-CDM, linked to AODB and METAR via SWIM/FIXM standards, no new system to learn
- Then other UAE airports and Gulf providers

1 · Users and customers *Who uses it? Who decides to adopt or pay for it? They may be different.*

- **Users:** approach and tower controllers, ATC supervisors
- Airport operations centre and stand planners
- Airline operations centres
- **Payers:** the air navigation provider (dans, under GCAA); airports as co-buyers
- Later: other UAE and Gulf airports and providers

8 · Costs *What will cost the most money, time or effort?*

- **Integration** with A-CDM, AODB and ATC systems (biggest effort)
- Safety assurance and certification; controller time during the pilot
- Hosting (cloud or on-premise), small LLM usage, ongoing maintenance

9 · Revenue and funding *Who pays, or what does it save? Not every solution sells something.*

- **Revenue:** annual licence per airport (SaaS) plus one-off integration and training fees
- **Funding:** GCAA / TDRA-sponsored pilot first
- **Savings:** less fuel burned holding, fewer diversions and repositioning, fewer missed connections, less CO_2. Same engine scales to other Gulf airports.

Can it be done? *What must exist for this to work?*

Yes. The data is open and exists today; our engine and panel already run on the real 3 Jan 2026 fog day. A pilot needs live API access and A-CDM / AODB integration.

Will it last? *Who keeps it running after the hackathon?*

Run by dans / GCAA under licence, with our team maintaining the engine. Every new weather event adds to the replay library, so it keeps improving.

Will it matter? *One number that would show impact.*

DXB fog arrivals averaged **86 min late vs 11 min** on normal days. On 3 Jan 2026 landings fell to **15/h vs 28**. DABAB aims to shrink that gap.

Biggest risk + first 3 steps *The guess that could break the model.*

Risk: controllers don't trust AI plans under pressure.

1. Replay 5 fog + 2 storm days with controllers
2. 30-day shadow mode at DXB
3. Go/no-go for a controlled pilot

AI Review

Accepted: optimizer plans with human approval; explanations grounded in optimizer output

Rejected: autonomous action without controller confirmation; AI landing pilot

To verify: real delay and fuel savings from replay; diversion counts (data under-records); paying customer with GCAA / dans

Data: Dubai Airports flight information (arrivals), Data.Dubai, May 2025 – Nov 2026; OMDB METAR via Iowa Environmental Mesonet; GCAA eAIP. Fog figures are observed averages over 5 fog events.